

EE 446: TinyML

Lab 4: Visual Wake Words

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**ELECTRICAL & COMPUTER
ENGINEERING**

UNIVERSITY *of* WASHINGTON



Lab #4

- Both Software and Hardware Lab!
- Develop a MobileNet model for Mask-Detection using **Edge Impulse**
- Using Transfer Learning to Train a Mask-Detection Model
- Compress MobileNet model for TinyML
- Deploy mask detection model in the Arduino board



Lab 4 – Introduction to Edge Impulse

Edge Impulse

- No Coding Required
- Edge Computing
- Support Various Sensors

The screenshot shows the Edge Impulse website homepage. At the top, a blue banner reads "Edge Impulse Optimizes NVIDIA AI for the Edge. [Read the Announcement!](#)". Below this is a navigation bar with the "EDGE IMPULSE" logo, and links for "Product", "Solutions", "Developers", "Pricing", "Company", "Blog", "Login", and a "Get started" button. The main content area features the headline "Build. Train. Optimize. AI for the edge." followed by a sub-headline: "Build datasets, train models, and optimize libraries to run on any edge device, from extremely low-power MCUs to efficient Linux CPU targets and GPUs." Below the text are two buttons: "Get Started" and "Schedule a demo". To the right of the text are three visual elements: a smartwatch displaying a "Heart Rate" graph, a "Sleep analysis" chart showing a 7h 32min duration, and a "Vibration" graph with a "Motion anomaly" alert. A large image of a robotic arm is also visible on the right side of the page.

Lab 4 [Software] – Edge Impulse Log In

1. Go to Website: <https://edgeimpulse.com/> ; Then Log In and create a new project

Log in



Username or e-mail

1

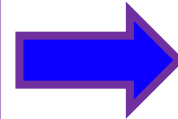


Password

[Forgot your password?](#)

Log in

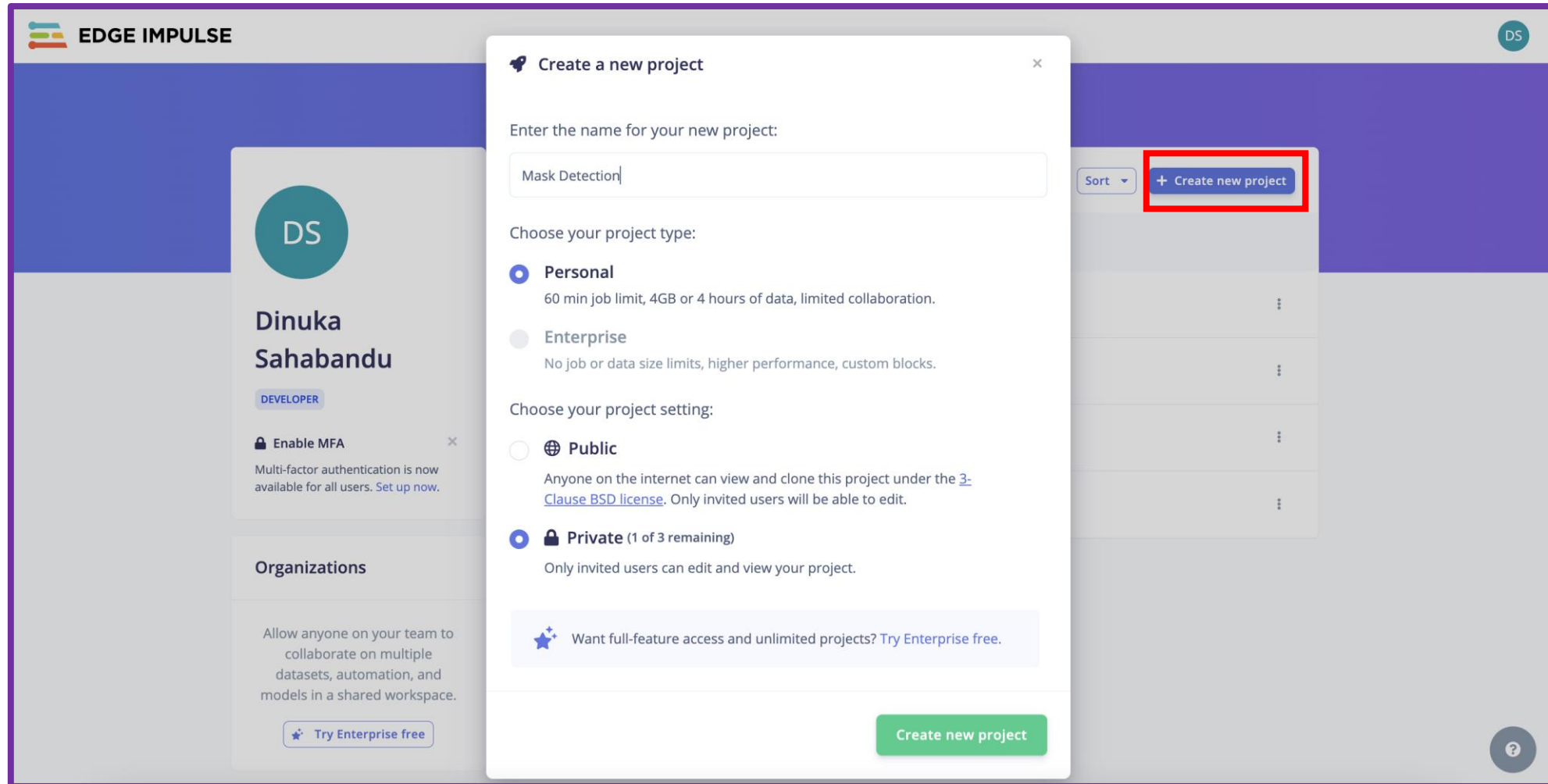
Don't have an account? [Sign up](#)



+ Create new project

Lab 4 [Software] – New Project

2. Create a New Project



Lab 4 [Software] – Choose Target Device

3. Scroll down and choose the target device first!

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Retrain model

Live classification

Model testing

Deployment

Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Project info

Mask

This is your Edge

+ New tag

Getting started

Start building

Add

Start with a

Not sure where

Target device

Define your target device requirements to inform model optimizations and performance calculations. No device yet? Use the default settings which you can change at any time.

Target device

Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz)

Processor family

Cortex-M

Clock rate

64

Max

Custom device name (optional)

Application budget

Specify the available RAM and ROM for the model's operation, along with the maximum allowed latency for your specific application. Not sure yet? Start with the defaults and modify them later on.

RAM

256

Max

ROM

1

Max

Latency

100

Max

Reset to default settings

Cancel

Save

Target: Cortex-M4F 80MHz

DS

Private

Published versions (0)

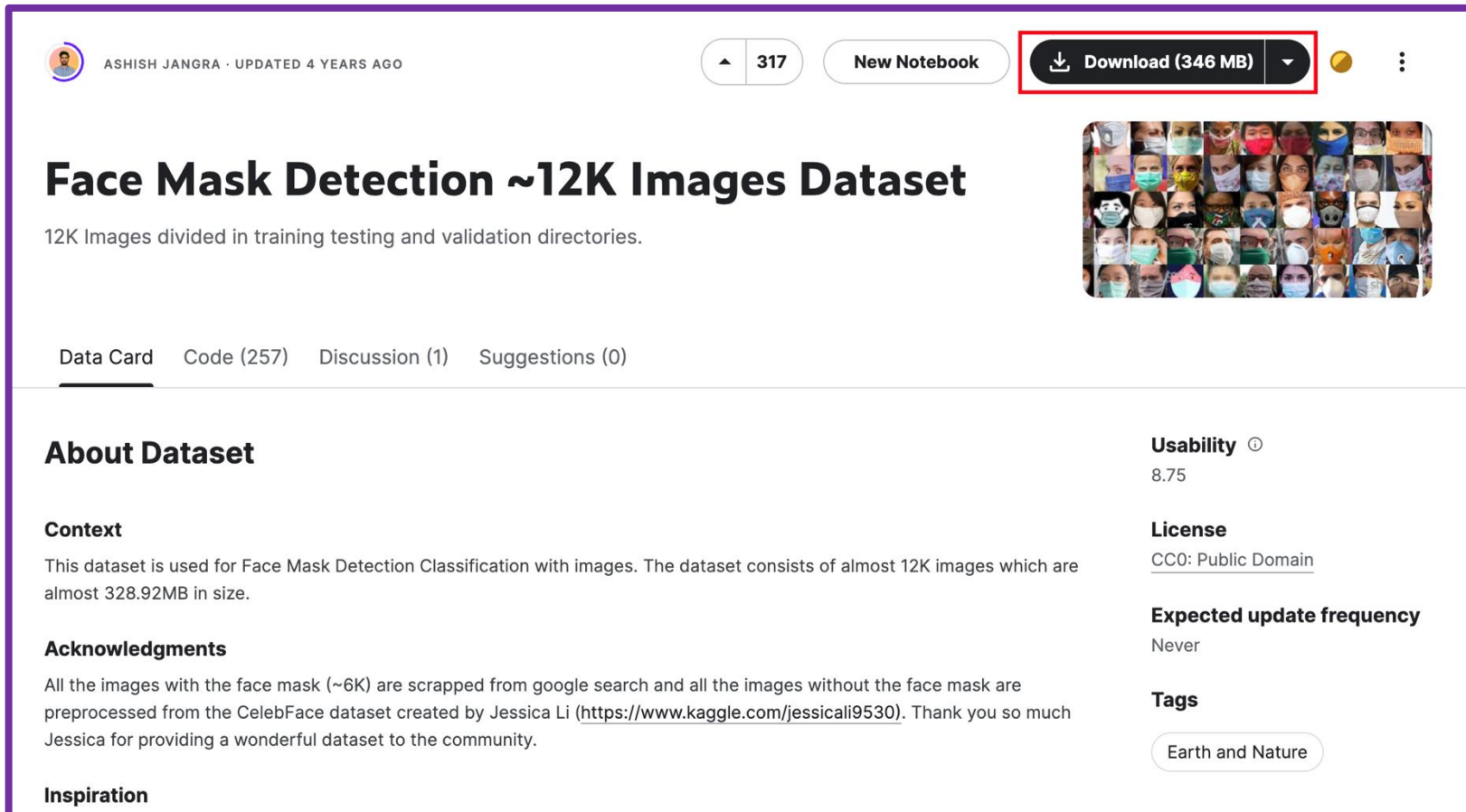
This project has no published versions.

Publish a version of your project



Lab 4 [Software] – Download Dataset

4. Download the dataset: We obtain a publicly available dataset from Kaggle
<https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset>



ASHISH JANGRA · UPDATED 4 YEARS AGO

317 New Notebook

Download (346 MB)

Face Mask Detection ~12K Images Dataset

12K Images divided in training testing and validation directories.

Data Card Code (257) Discussion (1) Suggestions (0)

About Dataset

Context

This dataset is used for Face Mask Detection Classification with images. The dataset consists of almost 12K images which are almost 328.92MB in size.

Acknowledgments

All the images with the face mask (~6K) are scrapped from google search and all the images without the face mask are preprocessed from the CelebFace dataset created by Jessica Li (<https://www.kaggle.com/jessicali9530>). Thank you so much Jessica for providing a wonderful dataset to the community.

Inspiration

Usability

8.75

License

CC0: Public Domain

Expected update frequency

Never

Tags

Earth and Nature

Lab 4 [Software] – Upload Dataset

5. Upload the dataset into your Edge Impulse project

In your Edge Impulse, click: “Add existing data”

The screenshot shows the Edge Impulse web interface for a project named "Mask Detection". The interface is divided into several sections:

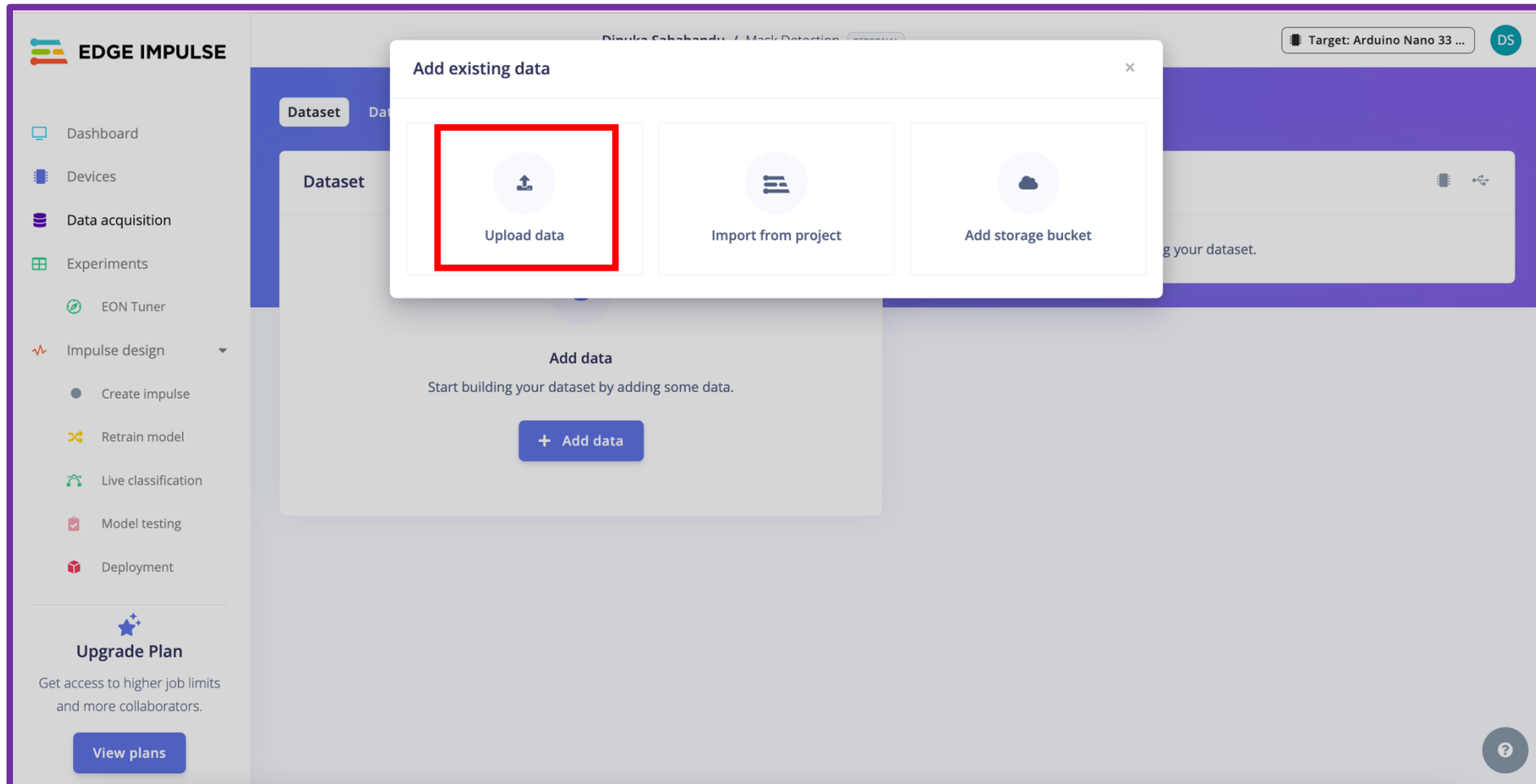
- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, and Impulse design. Under Impulse design, there are links for Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom, there is an "Upgrade Plan" section with a "View plans" button.
- Top Bar:** Displays the user name "Dinuka Sahabandu", the project name "Mask Detection", a "PERSONAL" tag, and a target device "Target: Arduino Nano 33 ...".
- Main Content Area:**
 - Project info:** The main header for the project, with tabs for Project info, Keys, Export, and Jobs.
 - Mask Detection:** The title of the project, followed by a description: "This is your Edge Impulse project. From here you acquire new training data, design impulses and train models." and a "+ New tag" button.
 - Getting started:** A section with the text "Start building your dataset or validate your model's on-device performance:". It contains three cards: "Add existing data" (highlighted with a red box), "Collect new data", and "Upload your model".
 - Start with a tutorial:** A section with the text "Not sure where to start? Follow a tutorial to build your first model in just minutes!".
 - Sharing:** A section with a "Private" dropdown menu and the text "This project is private, only invited users can edit and view."
 - Published versions (0):** A section with the text "This project has no published versions." and a "Publish a version of your project" button.

The URL at the bottom of the browser window is <https://studio.edgeimpulse.com/studio/739231/acquisition/training>.

Lab 4 [Software] – Upload Dataset

6. Upload the dataset into your Edge Impulse project

In your Edge Impulse, click: “Upload data”



Lab 4 [Software] – Upload Training Data Without Mask

7. Upload the “Training Data Without Mask” into your Edge Impulse project

Enter Label: “0”

The screenshot shows the 'Upload data' dialog in the Edge Impulse web interface. The left sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, and Upgrade Plan. The main content area of the dialog includes instructions on supported file formats (CBOR, JSON, CSV, Parquet, WAV, JPG, PNG, AVI, MP4) and an option to use the Python SDK. Below this, the 'Upload mode' section has two radio buttons: 'Select individual files' and 'Select a folder' (highlighted with a red box and labeled 'Step 1'). The 'Select files' section features a 'Choose Files' button (highlighted with a red box and labeled 'Step 4') and a 'No file chosen' status. The 'Upload into category' section has two radio buttons: 'Automatically split between training and testing' and 'Training' (highlighted with a red box and labeled 'Step 2'). The 'Label' section has two radio buttons: 'Infer from filename' and 'Leave data unlabeled', followed by an 'Enter label:' input field containing the value '0' (highlighted with a red box and labeled 'Step 3').

EDGE IMPULSE

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Get access to higher job limits and more collaborators.

View plans

Upload data

You can upload CBOR, JSON, CSV, Parquet, WAV, JPG, PNG, AVI or MP4 files. You can also upload an annotation file named "info.labels" with your data to assign bounding boxes, labels, and/or metadata. View [Uploader docs](#) to learn more. Alternatively, you can use our [Python SDK](#) to programmatically ingest data in various formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

☐ Select individual files ?

☒ Select a folder ? **Step 1**

Select files

Choose Files No file chosen **Step 4**

Upload into category

☐ Automatically split between training and testing ?

☒ Training **Step 2**

☐ Testing

Label

☐ Infer from filename ?

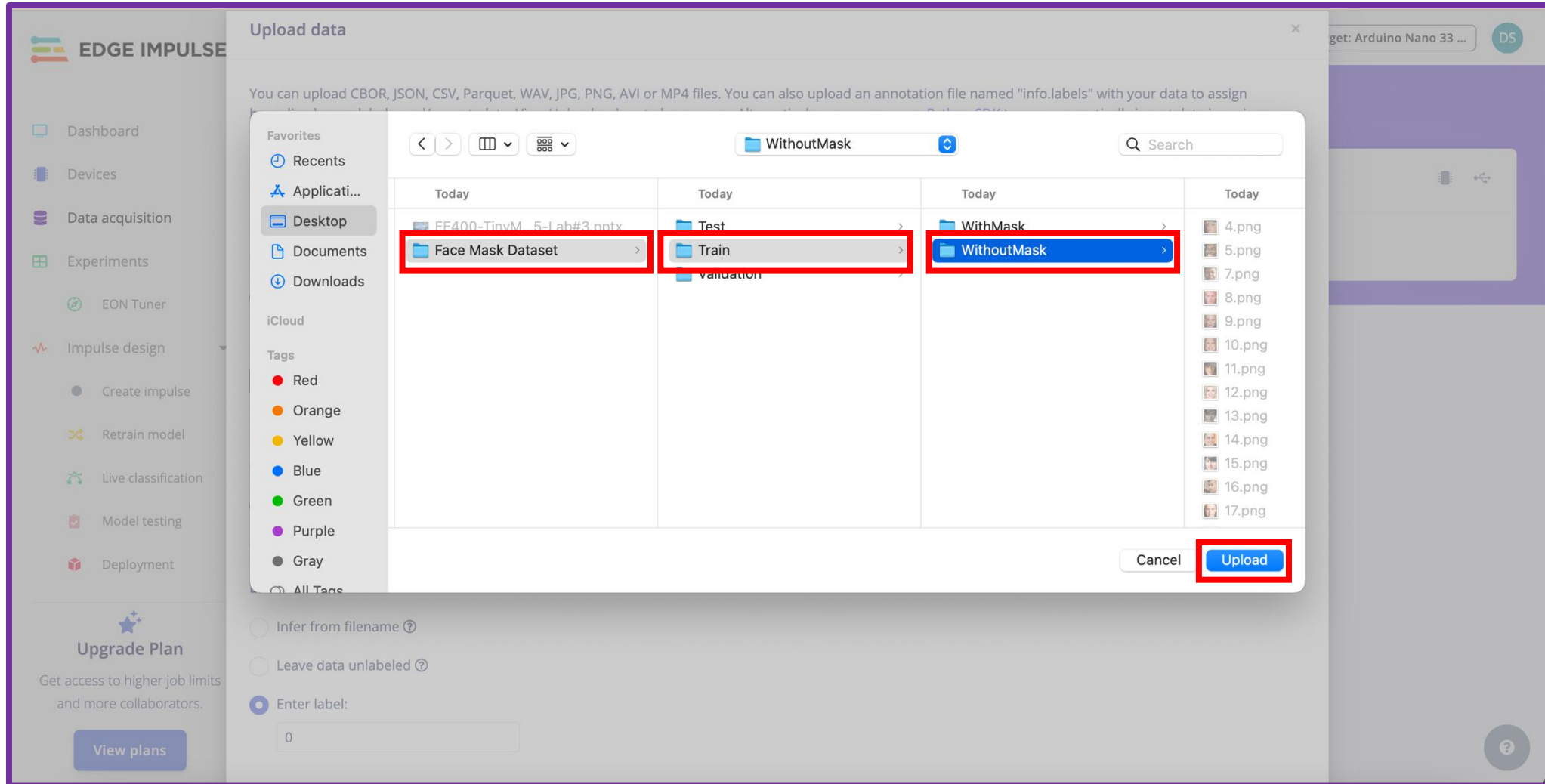
☐ Leave data unlabeled ?

☒ Enter label:

0 **Step 3**

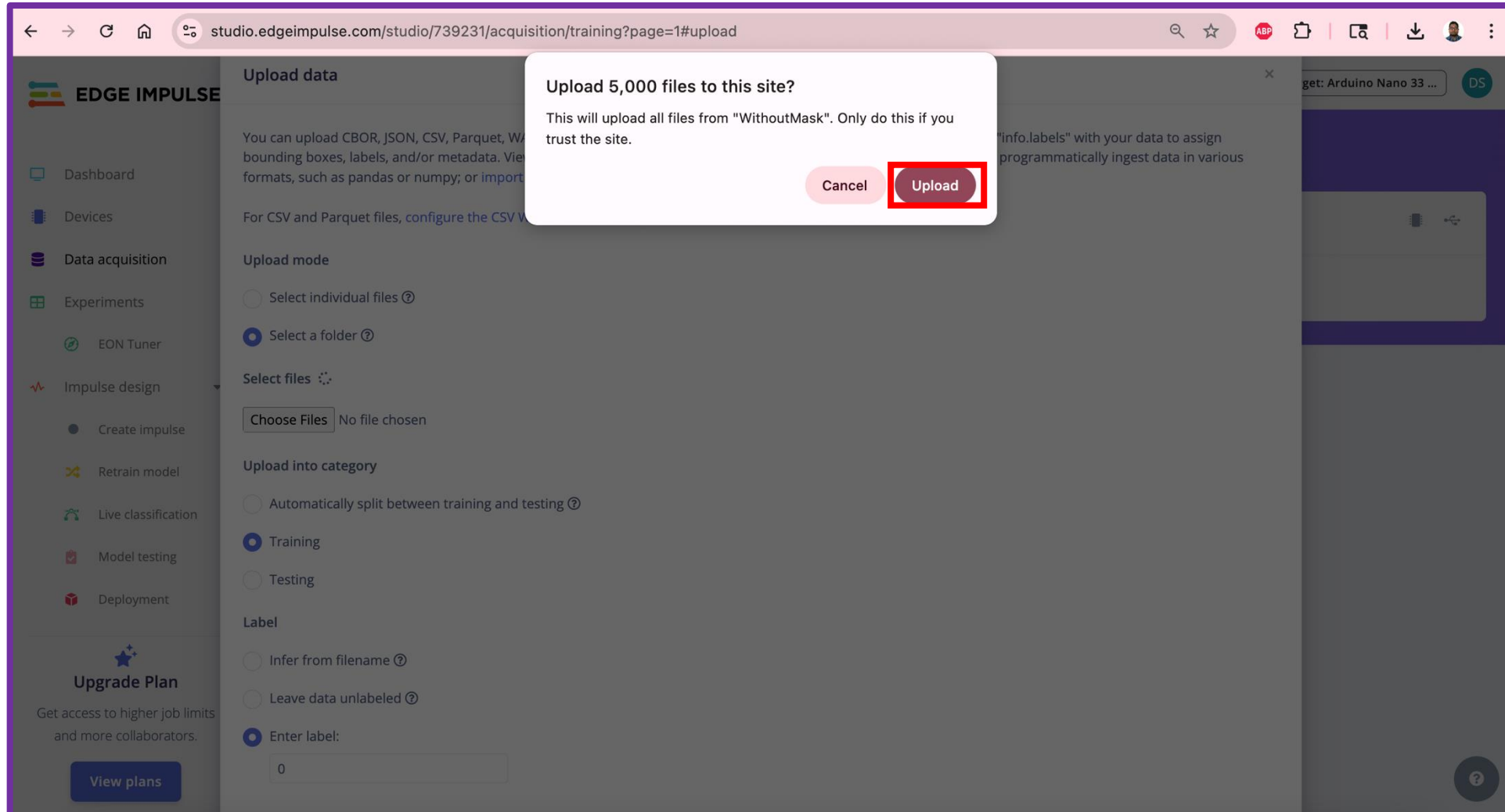
Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project



Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project



Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, Live classification, Model testing, and Deployment. Below these is an 'Upgrade Plan' section. The main content area shows the 'Upload data' workflow. It includes instructions on file formats and a link to the CSV Wizard. Under 'Upload mode', 'Select a folder' is selected. The 'Select files' section has a 'Choose Files' button and a limit of 5000 files. Under 'Upload into category', 'Training' is selected. The 'Label' section has 'Enter label:' selected, with a text input field containing the value '0'. At the bottom right, the 'Upload data' button is highlighted with a red rectangle. A 'Back' link is at the bottom left of the main area.

EDGE IMPULSE

formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

☐ Select individual files ?

☒ Select a folder ?

Select files

[Choose Files](#) 5000 files

Upload into category

☐ Automatically split between training and testing ?

☒ Training

☐ Testing

Label

☐ Infer from filename ?

☐ Leave data unlabeled ?

☒ Enter label:

0

[Back](#)

[Upload data](#)

Lab 4 [Software] – Upload Training Data Without Mask

8. Upload the “Training Data Without Mask” into your Edge Impulse project

EDGE IMPULSE

formats, such as pandas or numpy; or [import data directly from another project](#).

For CSV and Parquet files, [configure the CSV Wizard](#) to define how your files should be processed before uploading files.

Upload mode

- ☐ Select individual files ?
- ☒ Select a folder ?

Select files

[Choose Files](#) 5000 files

Upload into category

- ☐ Automatically split between training and testing
- ☒ Training
- ☐ Testing

Label

- ☐ Infer from filename ?
- ☐ Leave data unlabeled ?
- ☒ Enter label:

[Back](#)

Upload output

```
[ 578/5000] Uploading 4177.png OK  
[ 579/5000] Uploading 5269.png OK  
[ 580/5000] Uploading 3618.png OK  
[ 581/5000] Uploading 4163.png OK  
[ 582/5000] Uploading 2506.png OK
```

**We are doing
“Image Classification”
NOT Object Detection**

IMAGE DATA DETECTED!

Are you building an object
detection project?

You can change this choice under "Dashboard > Labeling method".

[Yes](#) [No](#)

[Upload data](#)

Lab 4 [Software] – Upload Training Data With Mask

9. Upload the “Training Data With Mask” into your Edge Impulse project Enter Label: “1”

The screenshot shows the Edge Impulse web interface with the following elements and annotations:

- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom is an Upgrade Plan section.
- Main Content Area:**
 - Instructions: "formats, such as pandas or numpy; or import data directly from another project." and "For CSV and Parquet files, configure the CSV Wizard to define how your files should be processed before uploading files."
 - Upload mode:** Two radio buttons: "Select individual files" and "Select a folder" (annotated with a red box and "Step 1").
 - Select files:** A "Choose Files" button next to "5000 files" (annotated with a red box and "Step 2").
 - Upload into category:** Two radio buttons: "Automatically split between training and testing" and "Training" (annotated with a red box and "Step 3").
 - Label:** Three radio buttons: "Infer from filename", "Leave data unlabeled", and "Enter label:" (annotated with a red box and "Step 4"). The "Enter label:" option has a text input field containing the number "1".
 - Upload output:** A list of file upload progress logs, each showing a file path and status (e.g., "[4983/5000] Uploading 586.png OK").
 - Bottom:** A "Back" link and an "Upload data" button (annotated with a red box and "Step 5").
- Annotations:** A purple callout box next to the "Training" radio button contains the text: "Face Mask Dataset", "→ 'Train'", and "→ 'WithMask'".

Lab 4 [Software] – Upload Test Data Without Mask

10. Upload the “**Test Data Without Mask**” into your Edge Impulse project
Enter Label: “0”

The screenshot shows the Edge Impulse web interface with a sidebar on the left containing navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Upgrade Plan, and View plans. The main content area is titled "For CSV and Parquet files, configure the CSV Wizard to define how your files should be processed before uploading files." and contains the following sections:

- Upload mode:** Two radio buttons are present. "Select individual files" is unselected, and "Select a folder" is selected and highlighted with a red box and labeled "Step 1".
- Select files:** A "Choose Files" button is highlighted with a red box and labeled "Step 2". It indicates "509 files" are selected.
- Upload into category:** Three radio buttons are present. "Automatically split between training and testing" is unselected. "Training" is unselected. "Testing" is selected and highlighted with a red box and labeled "Step 3".
- Label:** Three radio buttons are present. "Infer from filename" is unselected. "Leave data unlabeled" is unselected. "Enter label:" is selected and highlighted with a red box and labeled "Step 4". Below it, a text input field contains the value "0".
- Upload output:** A list of upload progress logs is shown, including file names and status (e.g., "Uploading Augmented_66_3706782.png OK"). A summary at the bottom states: "Done. Files uploaded successful: 4869. Files that failed to upload: 131."
- Upload data:** A blue button at the bottom right is highlighted with a red box and labeled "Step 5".

A purple callout box with a white background is overlaid on the "Testing" step, containing the text: "Face Mask Dataset" → "Test" → "WithoutMask".

Lab 4 [Software] – Upload Test Data With Mask

11. Upload the “Test Data With Mask” into your Edge Impulse project

Enter Label: “1”

The screenshot shows the Edge Impulse web interface with a sidebar on the left containing navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Upgrade Plan, and View plans. The main content area displays the 'Upload data' workflow. The interface is annotated with five steps:

- Step 1:** The 'Select a folder' radio button under 'Upload mode' is selected and highlighted with a red box.
- Step 2:** The 'Choose Files' button, which shows '483 files', is highlighted with a red box.
- Step 3:** The 'Testing' radio button under 'Upload into category' is selected and highlighted with a red box. A purple callout box next to it contains the text:
“Face Mask Dataset”
→ “Test”
→ “WithMask”
- Step 4:** The 'Enter label:' text input field, which contains the number '1', is highlighted with a red box.
- Step 5:** The 'Upload data' button at the bottom right is highlighted with a red box.

The 'Upload output' section on the right shows a list of files being uploaded (e.g., 4795.png, 2482.png, etc.) with status 'OK'. At the bottom, it states: 'Done. Files uploaded successful: 509. Files that failed to upload: 0. Job completed'.

Lab 4 [Software] – Observe Data

12. Observe uploaded data samples via “Data acquisition” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface for a project named 'Mask Detection'. The 'Data acquisition' tab is selected in the left sidebar. The main content area shows the following:

- Dataset Summary:** DATA COLLECTED 10,839 items. TRAIN / TEST SPLIT 91% / 9%.
- Dataset Table:** A table with columns: SAMPLE NAME, LABEL, and ADDED. It lists training and test samples.
- Sample Preview:** A preview of a data sample labeled 'Augmented_678_6388399' showing a person wearing a mask.
- Metadata:** A section for metadata, currently showing 'No metadata'.

A blue callout box highlights the 'TRAIN / TEST SPLIT 91% / 9%' and contains the text: 'Current Train/Test split is 91% / 9%. Next, we are going to change this to 80% / 20% by moving some samples from Training set to Test set'.

SAMPLE NAME	LABEL	ADDED
Augmented_383_82335	1	Today, 01:46:31
Augmented_650_1052629	1	Today, 01:46:31
Augmented_693_6192334	1	Today, 01:46:31
Augmented_363_2038261	1	Today, 01:46:31
Augmented_811_5146977	1	Today, 01:46:31
Augmented_491_5916085	1	Today, 01:46:31
Augmented_678_6388399	1	Today, 01:46:31
Augmented_50_771835	1	Today, 01:46:31

Lab 4 [Software] – Train/Test Split

13. Set Train/Test split via “Dashboard” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface. On the left sidebar, the 'Dashboard' tab is highlighted with a red rectangle. The main content area is divided into two sections. The top section, 'Administrative zone', contains a checkbox for 'Use xxd instead of INCBIN to link TFLite/ONNX files' and a 'Save performance settings' button. The bottom section, 'Danger zone', is highlighted with a blue wavy border and contains a red-bordered button labeled 'Perform train / test split'. A blue callout box with the text 'Scroll down to “Danger zone” section' points to this button. Other buttons in the 'Danger zone' include 'Launch getting started tutorial', 'Delete this project', and 'Delete all data in this project'. The footer shows '© 2025 EdgeImpulse Inc. All rights reserved' and a help icon.

EDGE IMPULSE

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Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Save performance settings

Administrative zone

☐ Use xxd instead of INCBIN to link TFLite/ONNX files

Save experiments

Danger zone

Perform train / test split

Launch getting started tutorial

Delete this project

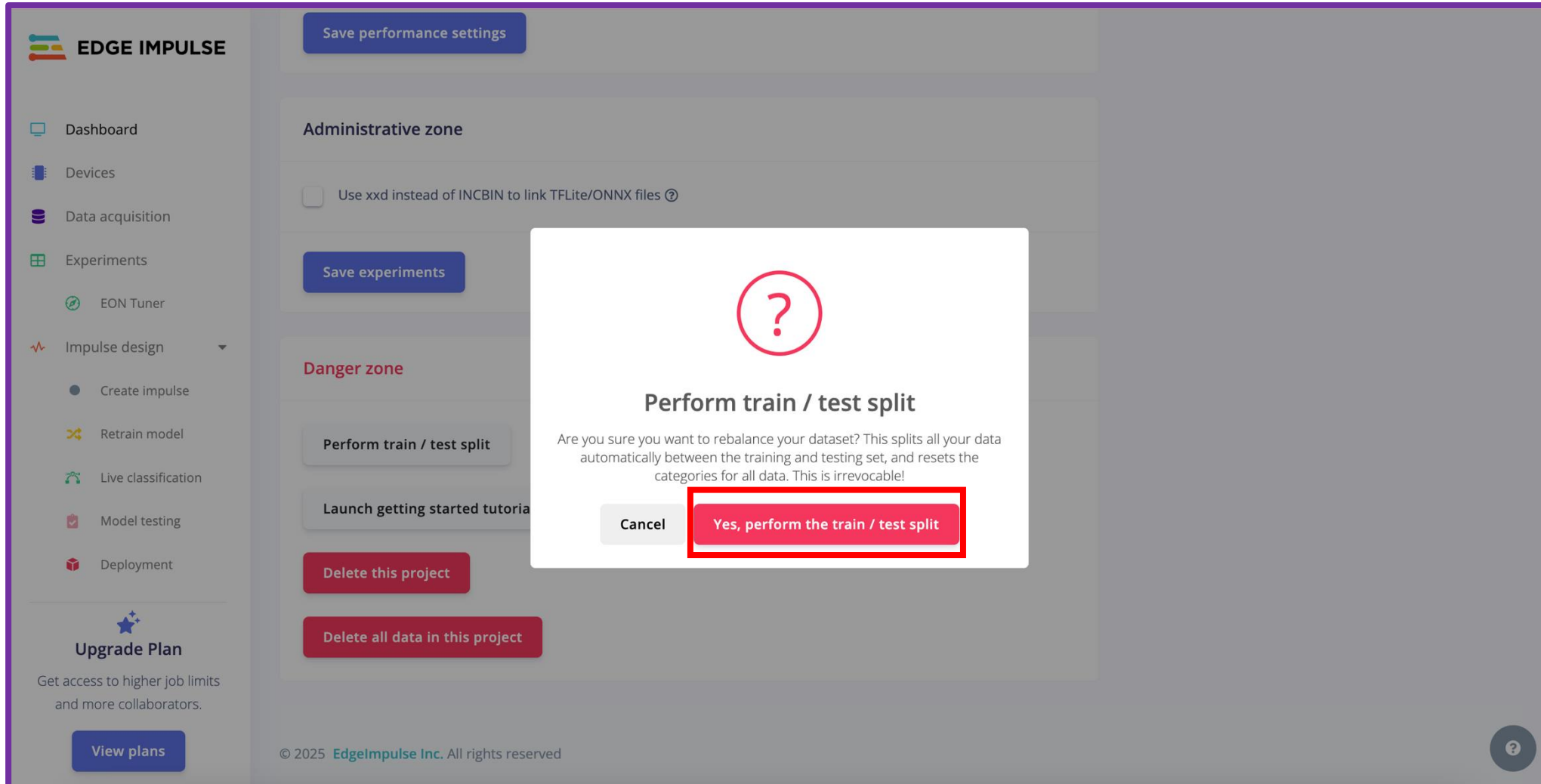
Delete all data in this project

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Scroll down to “Danger zone” section

Lab 4 [Software] – Train/Test Split

13. Set Train/Test split via “Dashboard” tab in your Edge Impulse project



Lab 4 [Software] – Train/Test Split

14. Observe updated Train/Test split via “Data acquisition” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface for a project named 'Mask Detection' by Dinuka Sahabandu. The 'Data acquisition' tab is selected in the left sidebar. The main content area shows a dataset with 10,839 items and a train/test split of 80% / 20%. A table lists training samples with their names, labels, and addition times.

SAMPLE NAME	LABEL	ADDED
Augmented_797_8118850	1	Today, 01:50:09
Augmented_673_1693056	1	Today, 01:50:09
Augmented_677_9933265	1	Today, 01:50:09
Augmented_744_2437969	1	Today, 01:50:09
351	1	Today, 01:50:09
423	1	Today, 01:50:08
Augmented_569_2915532	1	Today, 01:50:08
Augmented_164_2235307	1	Today, 01:50:08

Lab 4 [Software] – Create Impulse

15. Setup the ML Pipeline via “Create Impulse” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface for a project named 'Mask Detection' by 'Dinuka Sahabandu'. The target device is 'Arduino Nano 33 ...'. The left sidebar shows navigation options: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Retrain model, Live classification, Model testing, and Deployment. The 'Impulse design' section is active, and the 'Create impulse' option is highlighted with a red box and labeled 'Step 1'.

The main workspace shows the 'Impulse #1' configuration. It includes a description: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.' The workflow consists of three main blocks: 'Image data' (red), 'Add a processing block' (light blue, outlined in red and labeled 'Step 3'), and 'Add a learning block' (light blue). The 'Image data' block is further configured with 'Input axes' set to 'image', 'Image width' and 'Image height' both set to '48' (highlighted with a blue box and labeled 'Step 2'), and 'Resize mode' set to 'Fit shortest axis'. A red arrow points from a text box 'Set Image width and height to 48 × 48' to the 'Image width' and 'Image height' input fields. The 'Output features' block is green and contains a checkmark. A 'Save Impulse' button is located at the bottom right of the main workspace.

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Lab 4 [Software] – Create Impulse

16. Add “Image” data processing block

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main area displays 'Impulse #1' with a red 'Image data' block. A modal dialog titled 'Add a processing block' is open in the center. It contains a table with two rows: 'Image' and 'Raw Data'. The 'Image' row is highlighted with a red rectangle. The 'Image' block description is 'Preprocess and normalize image data, and optionally reduce the color depth.' The 'Raw Data' block description is 'Use data without pre-processing. Useful if you want to use deep learning to learn features.' Both blocks are by 'Edge Impulse' and have an 'Add' button. Below the table, there is a note: 'Some processing blocks have been hidden based on the data in your project. Show all blocks anyway'. At the bottom of the dialog are 'Add custom block' and 'Cancel' buttons. The background interface shows the 'Input axes' section with 'Image width' set to 48 and 'Resize mode' set to 'Fit shortest'. On the right, there is an 'Output features' section with a checkmark and a 'Save Impulse' button. The top right corner shows the target device 'Target: Arduino Nano 33 ...' and a user profile icon 'DS'.

DESCRIPTION	AUTHOR	RECOMMENDED
Image OFFICIALLY SUPPORTED Preprocess and normalize image data, and optionally reduce the color depth.	Edge Impulse	★ Add
Raw Data OFFICIALLY SUPPORTED Use data without pre-processing. Useful if you want to use deep learning to learn features.	Edge Impulse	Add

Some processing blocks have been hidden based on the data in your project. [Show all blocks anyway](#)

[Add custom block](#) [Cancel](#)

Lab 4 [Software] – Create Impulse

17. Click on “Add a learning block”

The screenshot shows the Edge Impulse web interface for configuring 'Impulse #1'. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom, there is an 'Upgrade Plan' section with a 'View plans' button.
- Header:** Displays 'Impulse #1' and a description: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.'
- Main Content Area:** Contains three primary blocks:
 - Image data (Red):** Includes 'Input axes' set to 'image', 'Image width' and 'Image height' both set to 48, and a 'Resize mode' dropdown set to 'Fit shortest axis'.
 - Image (White):** Includes a 'Name' field set to 'Image', 'Input axes (1)' set to 'Image', and a dropdown menu set to 'image'.
 - Add a learning block (Red dashed box):** A button with a flask icon and the text 'Add a learning block'.
- Output features (Teal):** A button with a checkmark icon and the text 'Output features'.
- Bottom:** A 'Save Impulse' button and a help icon (question mark).

Lab 4 [Software] – Create Impulse

18. Add “Transfer Learning (Images)” as your learning block

The screenshot shows the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. The main area displays 'Impulse #1' with a red sidebar for 'Image data' containing 'Input axes' (image), 'Image width' (48), and 'Resize mode' (Fit shortest). A modal window titled 'Add a learning block' is open in the center. It contains a table of learning blocks:

DESCRIPTION	AUTHOR	RECOMMENDED
Classification OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for categorizing movement or recognizing audio.	Edge Impulse	<button>Add</button>
Transfer Learning (Images) OFFICIALLY SUPPORTED Fine tune a pre-trained image classification model on your data. Good performance even with relatively small image datasets.	Edge Impulse	<button>Add</button>
Regression OFFICIALLY SUPPORTED Learns patterns from data, and can apply these to new data. Great for predicting numeric continuous values.	Edge Impulse	<button>Add</button>
Visual Anomaly Detection - FOMO-AD OFFICIALLY SUPPORTED Detect visual anomalies. Extracts visual features using a pre-trained backbone, and applies a scoring function to evaluate how anomalous a sample is by comparing the extracted features to the learned model. Does not require anomalous data.	Edge Impulse	<button>Add</button>

Below the table is a lightning bolt icon and the text: 'Want access to all learning blocks? Upgrade now.' At the bottom of the modal, it says: 'Some learning blocks have been hidden based on the data in your project. Show all blocks anyway'. In the background, the 'Output features' section is visible with a green box and a checkmark, and a 'Save Impulse' button.

Lab 4 [Software] – Save Impulse

19. Click on “Save Impulse” to save your ML Pipeline

The screenshot displays the Edge Impulse web interface for configuring an ML pipeline. The interface is divided into a left sidebar, a top header, and a main workspace.

- Left Sidebar:** Contains navigation links for Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Retrain model, Live classification, Model testing, and Deployment. At the bottom, there is an "Upgrade Plan" section with a "View plans" button.
- Top Header:** Labeled "Impulse #1", it includes a description: "An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data."
- Main Workspace:** Features four colored blocks representing different stages of the pipeline:
 - Image data (Red):** Configured with "Input axes" set to "image", "Image width" and "Image height" both set to 48, and "Resize mode" set to "Fit shortest axis".
 - Image (White):** Configured with "Name" set to "Image" and "Input axes (1)" set to "image".
 - Transfer Learning (Images) (Blue):** Configured with "Name" set to "Transfer learning", "Input features" checked for "Image", and "Output features" set to "2 (0, 1)".
 - Output features (Green):** Configured with "Output features" set to "2 (0, 1)".
- Bottom of Workspace:** Includes two dashed boxes for "Add a processing block" and "Add a learning block", and a "Save Impulse" button highlighted with a red rectangle.

Lab 4 [Software] – Setup Image Data

20. Setup the image data via “Image” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left sidebar, the 'Image' option is selected and highlighted with a red box. The top navigation bar shows the 'Generate features' button highlighted with a red box. The main content area is divided into several sections:

- Raw data:** Displays a sample image of a person wearing a green face mask. It includes filters for 'Show: All labels' and 'Augmented_797_811'.
- Raw features:** Shows a list of hexadecimal values representing raw image data: 0x5a5d52, 0x5a5d52, 0x585b51, 0x575a50, 0x56594f, 0x55584f, 0x55584f, 0x55584f...
- Parameters:** Includes a section for 'Image' with a 'Color depth' dropdown menu set to 'RGB'. A 'Save parameters' button is located at the bottom of this section.
- DSP result:** Displays the processed image of the person wearing the green face mask.
- Processed features:** Shows a list of numerical values representing processed image data: 0.3529, 0.3647, 0.3216, 0.3529, 0.3647, 0.3216, 0.3451, 0.3569, 0.3176, 0.3412...
- On-device performance:** A section with a question mark icon, indicating performance metrics.

Lab 4 [Software] – Setup Image Data

20. Setup the image data via “Image” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface for a project named "Mask Detection" by Dinuka Sahabandu. The interface is divided into a left sidebar, a main content area, and a top header. The sidebar contains navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning, Retrain model, and Live classification. The main content area has two tabs: "Parameters" and "Generate features" (active). Under "Parameters", the "Training set" section shows "Data in training set" as 8,622 items and "Classes" as 2 (0, 1). A red rectangle highlights the "Generate features" button. Below this is the "Feature generation output" section, which currently shows 0 items. The "Feature explorer" section on the right states "No features generated yet." The top header shows the target device as "Target: Arduino Nano 33 ..." and a user profile icon labeled "DS". A footer at the bottom left of the interface says "© 2025 EdgeImpulse Inc. All rights reserved."

EDGE IMPULSE

Dinuka Sahabandu / Mask Detection PERSONAL

Target: Arduino Nano 33 ... DS

Parameters **Generate features**

Training set

Data in training set	8,622 items
Classes	2 (0, 1)

Generate features

Feature explorer

No features generated yet.

Feature generation output 0

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Upgrade Plan

Get access to higher job limits and more collaborators.

View plans

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning (highlighted with a red box), Retrain model, and Live classification. Below these is an 'Upgrade Plan' section. The main content area is titled 'Neural Network settings' and contains two sections: 'Training settings' and 'Advanced training settings'. The 'Training settings' section includes fields for 'Number of training cycles' (set to 20), 'Use learned optimizer' (checkbox), 'Learning rate' (set to 0.0005), 'Training processor' (set to CPU), and 'Data augmentation' (checkbox). The 'Advanced training settings' section is currently collapsed. Below these settings is the 'Neural network architecture' section, which shows an 'Input layer (27,648 features)' and an 'Output layer (2 classes)'. In the center of the architecture is a box for the selected model, 'MobileNetV2 96x96 0.35 (final layer: 16 neurons, 0.1 dropou', with icons for edit, expand, and delete. Below this model box is a red-outlined button labeled 'Choose a different model'. On the right side of the interface is a 'Training output' panel, which is currently empty and shows '(0)' results.

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

MobileNetV1 96x96 0.2 OFFICIALLY SUPPORTED Uses around 83.1K RAM and 218.3K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV1 96x96 0.1 OFFICIALLY SUPPORTED Uses around 53.2K RAM and 101K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.35 OFFICIALLY SUPPORTED Uses around 296.8K RAM and 575.2K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.1 OFFICIALLY SUPPORTED Uses around 270.2K RAM and 212.3K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 96x96 0.05 OFFICIALLY SUPPORTED Uses around 265.3K RAM and 162.4K ROM with default settings and optimizations. Works best with 96x96 input size. Supports both RGB and grayscale.	Edge Impulse	Add
MobileNetV2 160x160 1.0 OFFICIALLY SUPPORTED Uses around 1.3M RAM and 2.6M ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add
MobileNetV2 160x160 0.75 OFFICIALLY SUPPORTED Uses around 1.3M RAM and 1.7M ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add
MobileNetV2 160x160 0.5 OFFICIALLY SUPPORTED Uses around 700.7K RAM and 982.4K ROM with default settings and optimizations. Works best with 160x160 input size. Supports RGB only.	Edge Impulse	Add

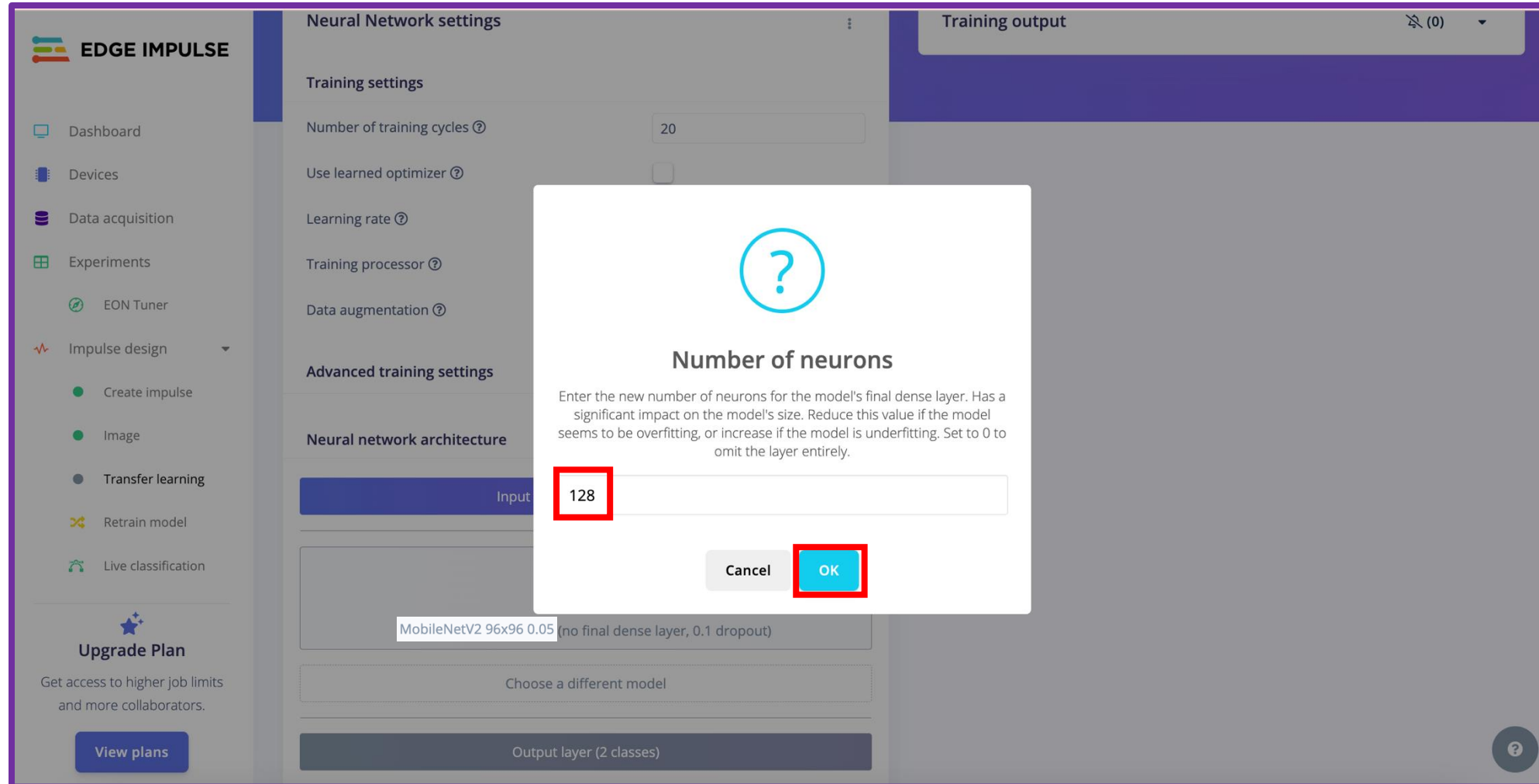
Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning (highlighted), Retrain model, and Live classification. Below these is an 'Upgrade Plan' section. The main content area is titled 'Neural Network settings' and includes 'Training settings' (Number of training cycles: 20, Use learned optimizer: unchecked, Learning rate: 0.0005, Training processor: CPU, Data augmentation: unchecked) and 'Advanced training settings' (collapsed). Under 'Neural network architecture', there is an 'Input layer (27,648 features)' and a highlighted 'MobileNetV2 96x96 0.05 (no final dense layer, 0.1 dropout)' layer. A red box and a red hexagon highlight this layer. Below it is a 'Choose a different model' button and an 'Output layer (2 classes)'. On the right, a 'Training output' section shows '(0)' results.

Lab 4 [Software] – Transfer Learning

21. Setup ML model via “Transfer Learning” tab in your Edge Impulse project



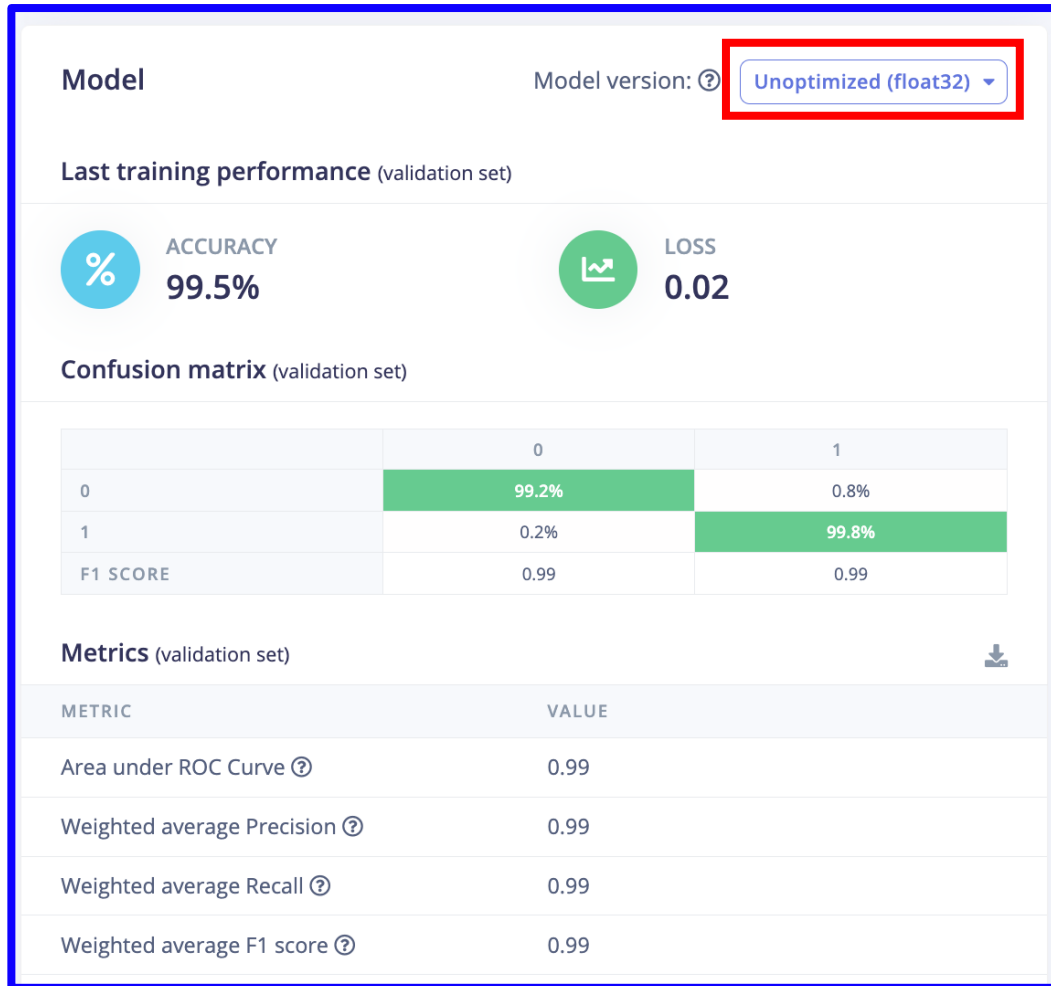
Lab 4 [Software] – Training via Transfer Learning

22. Select “Save & Train” to perform transfer learning on your ML model

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning, Retrain model, and Live classification. Below these is an 'Upgrade Plan' section. The main area is titled 'EDGE IMPULSE' and contains training configuration options: 'Number of training cycles' (20), 'Use learned optimizer' (checkbox), 'Learning rate' (0.0005), 'Training processor' (CPU), and 'Data augmentation' (checkbox). An 'Advanced training settings' dropdown is also present. The 'Neural network architecture' section shows an 'Input layer (27,648 features)', a central box for 'MobileNetV2 96x96 0.05 (final layer: 128 neurons, 0.1 dropout)' with a 'Choose a different model' link, and an 'Output layer (2 classes)'. At the bottom right, the 'Save & train' button is highlighted with a red rectangle. A help icon (?) is in the bottom right corner.

Lab 4 [Software] – Training Performance on Base Model

23. Observe the training performance of your base (unoptimized) model



Lab 4 [Software] – Training Performance on Compressed Model

24. Observe the training performance of your compressed (int8 quantized/optimized) model

Model Model version: ③ **Quantized (int8)** ▼

Last training performance (validation set)

ACCURACY 99.2% **LOSS** 0.04

Confusion matrix (validation set)

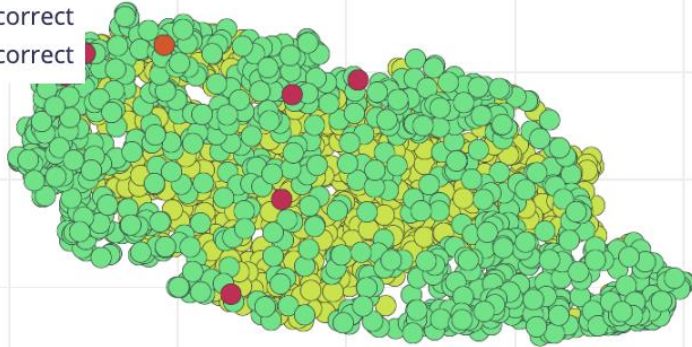
	0	1
0	98.9%	1.1%
1	0.4%	99.6%
F1 SCORE	0.99	0.99

Metrics (validation set) ⬇

METRIC	VALUE
Area under ROC Curve ③	0.99
Weighted average Precision ③	0.99
Weighted average Recall ③	0.99
Weighted average F1 score ③	0.99

Data explorer (full training set) ③

● 0 - correct
● 1 - correct
● 0 - incorrect
● 1 - incorrect



On-device performance ③ **TensorFlow Lite™ Compiler (RAM optimized)** ▼

INFERENCING TIME 829 ms. **PEAK RAM USAGE** 134.3K **FLASH USAGE** 328.8K

This is much better! (Less than 250K)

Lab 4 [Software] – Test Performance

25. Evaluate your unoptimized and optimized models on test data (data not used during the training – unseen data) via “**Model Testing**” tab in your Edge Impulse project

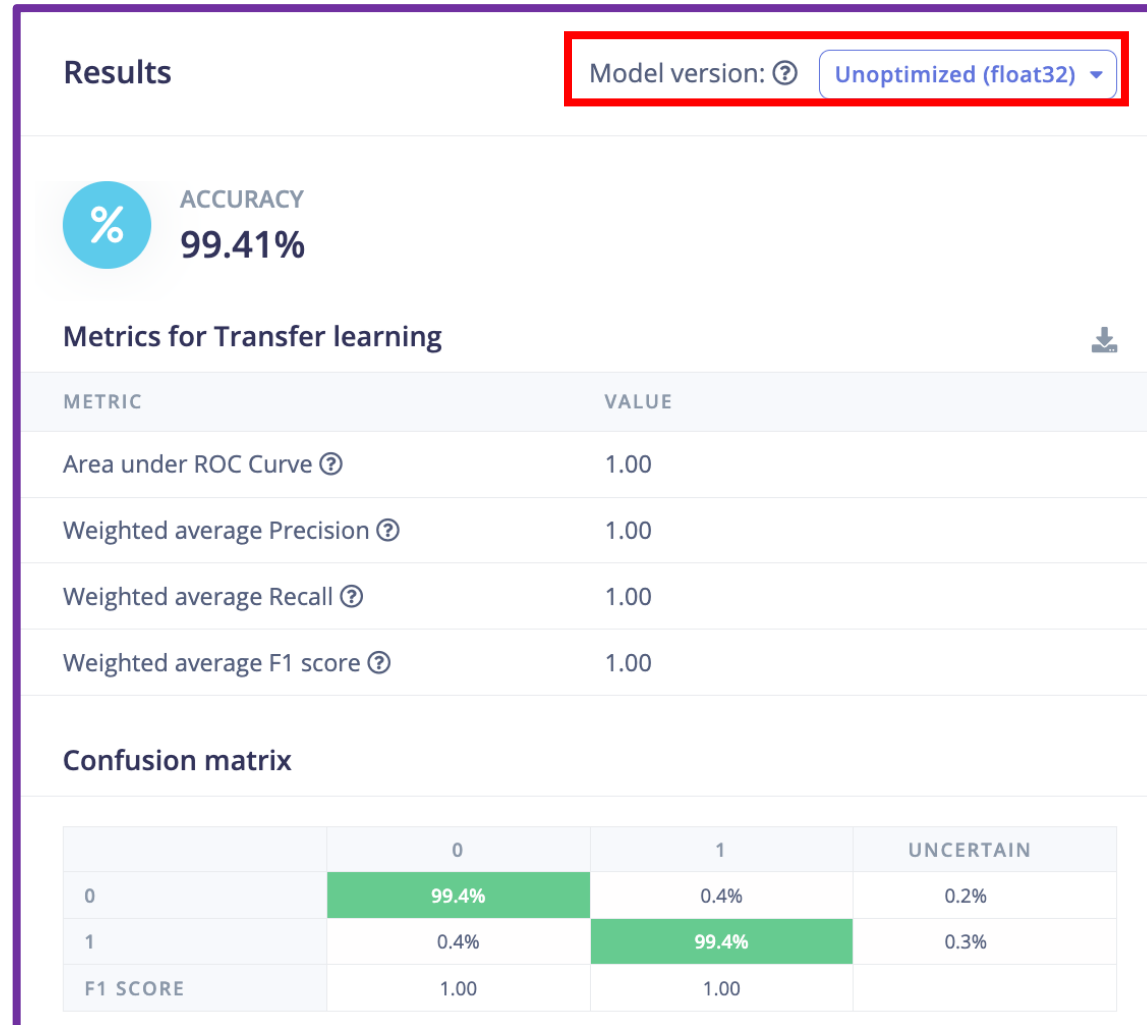
The screenshot displays the Edge Impulse web interface for a project named 'Mask Detection' by 'Dinuka Sahabandu'. The left sidebar contains navigation options: Devices, Data acquisition, Experiments, EON Tuner, Impulse design (selected), Create impulse, Image, Transfer learning, Retrain model, Live classification, **Model testing** (highlighted), Deployment, and Versioning. An 'Upgrade Plan' button is also visible. The main content area is divided into two panels. The left panel, titled 'Test data', contains a table with the following data:

SAMPLE NAME	EXPECTED OUTCOME	ACCURACY	RESULT
Augmented_7...	1		...
Augmented_1...	1		...
Augmented_3...	1		...
Augmented_6...	1		...
Augmented_1...	1		...
Augmented_6...	1		...
Augmented_2...	1		...
Augmented_7...	1		...

The right panel, titled 'Model testing output', shows a filter icon and '(0)' results. A 'Classify all' button is highlighted in the top right of the 'Test data' panel.

Lab 4 [Software] – Test Performance on Base Model

23. Observe the test performance (on unseen data) of your base (unoptimized) model



Lab 4 [Software] – Training Performance on Compressed Model

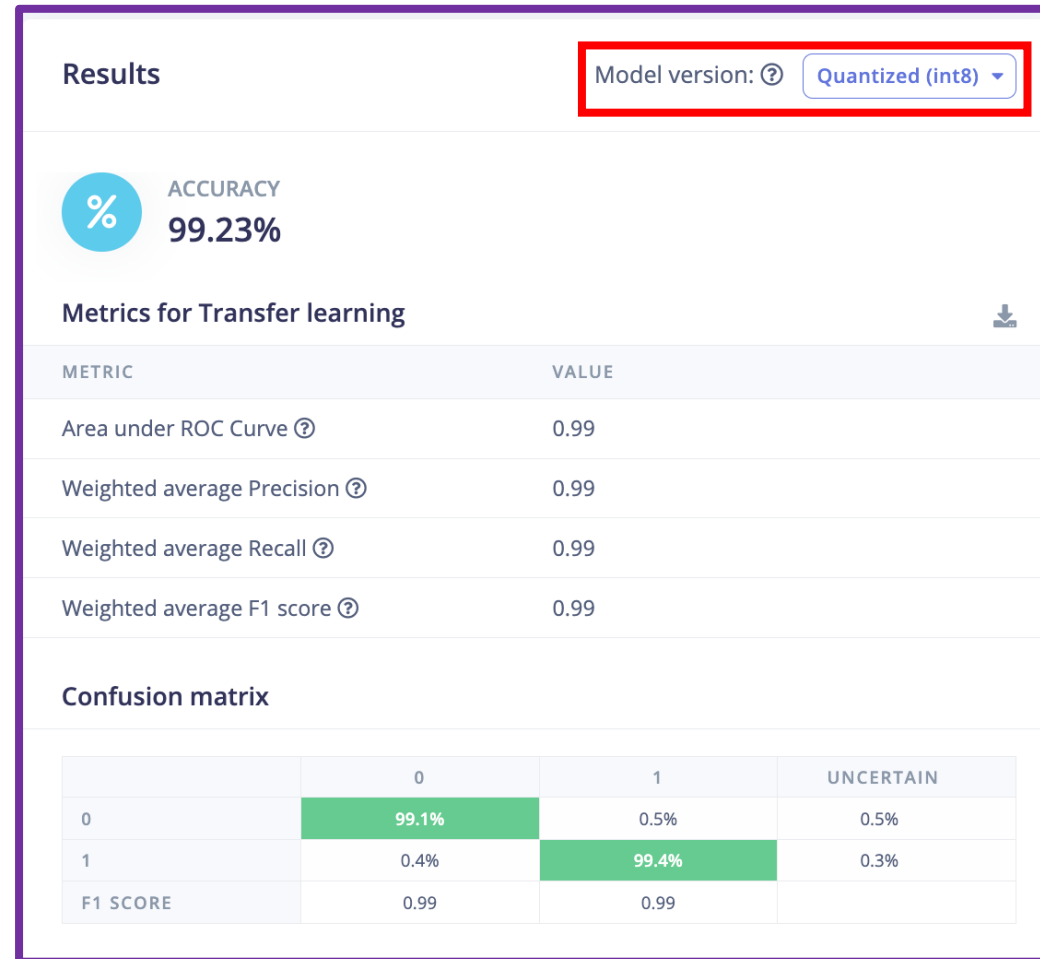
24. Observe the test performance (on unseen data) of your compressed (int8 quantized/optimized) model

The screenshot displays the AWS SageMaker console interface. On the left is a navigation sidebar with options: Devices, Data acquisition, Experiments, EON Tuner, Impulse design, Create impulse, Image, Transfer learning, Retrain model, Live classification, Model testing, Deployment, and Versioning. The main area is titled 'Test data' and contains a table of test results. A red box highlights the 'Classify all' button and a gear icon. A blue box highlights the 'Test 'Quantized (int8)' model' entry in the 'MODEL VARIANTS' section. A white box labeled 'Step 2' points to the 'Test 'Quantized (int8)' model' entry. A blue box labeled 'Step 1' points to the 'Classify all' button. A blue box labeled 'Step 3' points to the 'Test 'Quantized (int8)' model' entry. A blue box labeled 'Click the gear icon and re-run "Classify all" to view results for your int8 mode' points to the gear icon. The 'Results' section shows 'Model version: Unoptimized (float32)' and 'ACCURACY 99.41%'. Below this is a table of 'Metrics for Transfer learning'.

METRIC	VALUE
Area under ROC Curve	1.00
Weighted average Precision	1.00
Weighted average Recall	1.00
Weighted average F1 score	1.00

Lab 4 [Software] – Training Performance on Compressed Model

25. Observe the test performance (on unseen data) of your compressed (int8 quantized/optimized) model



Lab 4 [Software] – Model Deployment

26. Deploying your int8 model via “Deployment” tab in your Edge Impulse project

The screenshot shows the Edge Impulse web interface. On the left sidebar, the 'Deployment' tab is highlighted with a red box. The main area is titled 'Configure your deployment' and contains a search bar with 'Arduino Nano 33 BLE Sense' entered. Below the search bar, a red box highlights the search result for 'Arduino Nano 33 BLE Sense', which is described as 'A binary containing both the Edge Impulse data acquisition client and your full impulse.' Below this, there is a section for 'MODEL OPTIMIZATIONS' and a table showing performance metrics for the 'Quantized (int8)' model.

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Search results:

- Arduino Nano 33 BLE Sense**
A binary containing both the Edge Impulse data acquisition client and your full impulse.

MODEL OPTIMIZATIONS

Model optimizations can increase on-device performance but may reduce accuracy.

Quantized (int8)

Selected ✓

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	829 ms.	840 ms.
RAM	4.0K	125.0K	125.0K
FLASH	-	316.9K	-
ACCURACY			99.23%

Run this model

Scan QR code or launch in browser to test your prototype

[Launch in browser](#)

Lab 4 [Software] – Model Deployment

27. Select “Build” to generate binaries of the int8 model for deployment

The screenshot displays the EON Tuner interface. On the left is a sidebar with navigation options: EON Tuner, Impulse design, Create impulse, Image, Transfer learning, Retrain model, Live classification, Model testing, Deployment, Versioning, GETTING STARTED, Documentation, Forums, Upgrade Plan, and View plans. The main area is titled 'MODEL OPTIMIZATIONS' and includes a warning: 'Model optimizations can increase on-device performance but may reduce accuracy.' Below this, the 'EON™ Compiler' is selected, showing 'Same accuracy, 18% less RAM, 19% less ROM.' Two tables compare 'Quantized (int8)' and 'Unoptimized (float32)' models. The 'Quantized (int8)' model is selected. A large blue 'Build' button is at the bottom.

MODEL OPTIMIZATIONS
Model optimizations can increase on-device performance but may reduce accuracy.

EON™ Compiler
Same accuracy, 18% less RAM, 19% less ROM.

Quantized (int8)
Selected ✓

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	829 ms.	840 ms.
RAM	4.0K	125.0K	125.0K
FLASH	-	316.9K	-
ACCURACY			99.23%

Unoptimized (float32)
Select

	IMAGE	TRANSFER LEARN...	TOTAL
LATENCY	11 ms.	8,208 ms.	8,219 ms.
RAM	4.0K	300.3K	300.3K
FLASH	-	985.7K	-
ACCURACY			99.41%

Estimate for Arduino Nano 33 BLE Sense (Cortex-M4F 64MHz).

Build

Lab 4 [Software] – Model Deployment

28. Upon successful model build you will see the following. You should be now able to see **“.zip” file** in your **“Downloads” folder** that contain **flash files** for model deployment. **Unzip this folder.**

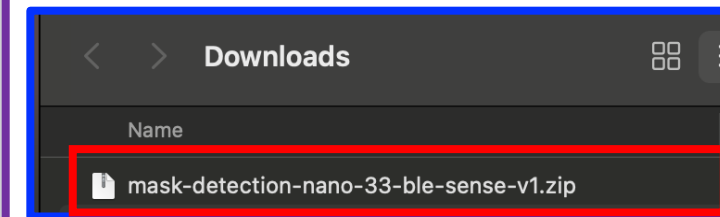
Built Arduino Nano 33 BLE Sense firmware

Click on the script for your operating system to flash the binary.

```
janjongboom@mac:~$ ./flash_mac.command
To update your account to use zsh, please run 'chsh -s /bin/zsh'.
For more details, please visit https://support.apple.com/kb/HT208050.
janjongboom@mac:~$ ./flash_mac.command
Finding Arduino Mbed core...
Finding Arduino Nano 33 BLE...
Finding Arduino Nano 33 BLE OK
Flashing board...
Device : nRF52840-QIAA
Version : Arduino Bootloader (SAM-BA extended) 2.0 [Arduino:IKXYZ]
Address : 0x0
Pages : 256
Page Size : 4096 bytes
Total Size : 1024KB
Planes : 1
Lock Regions : 0
Locked : none
Security : false
Erase flash
Done in 0.000 seconds
Write 527696 bytes to flash (129 pages)
[=====] 23% (30/129 pages)
```

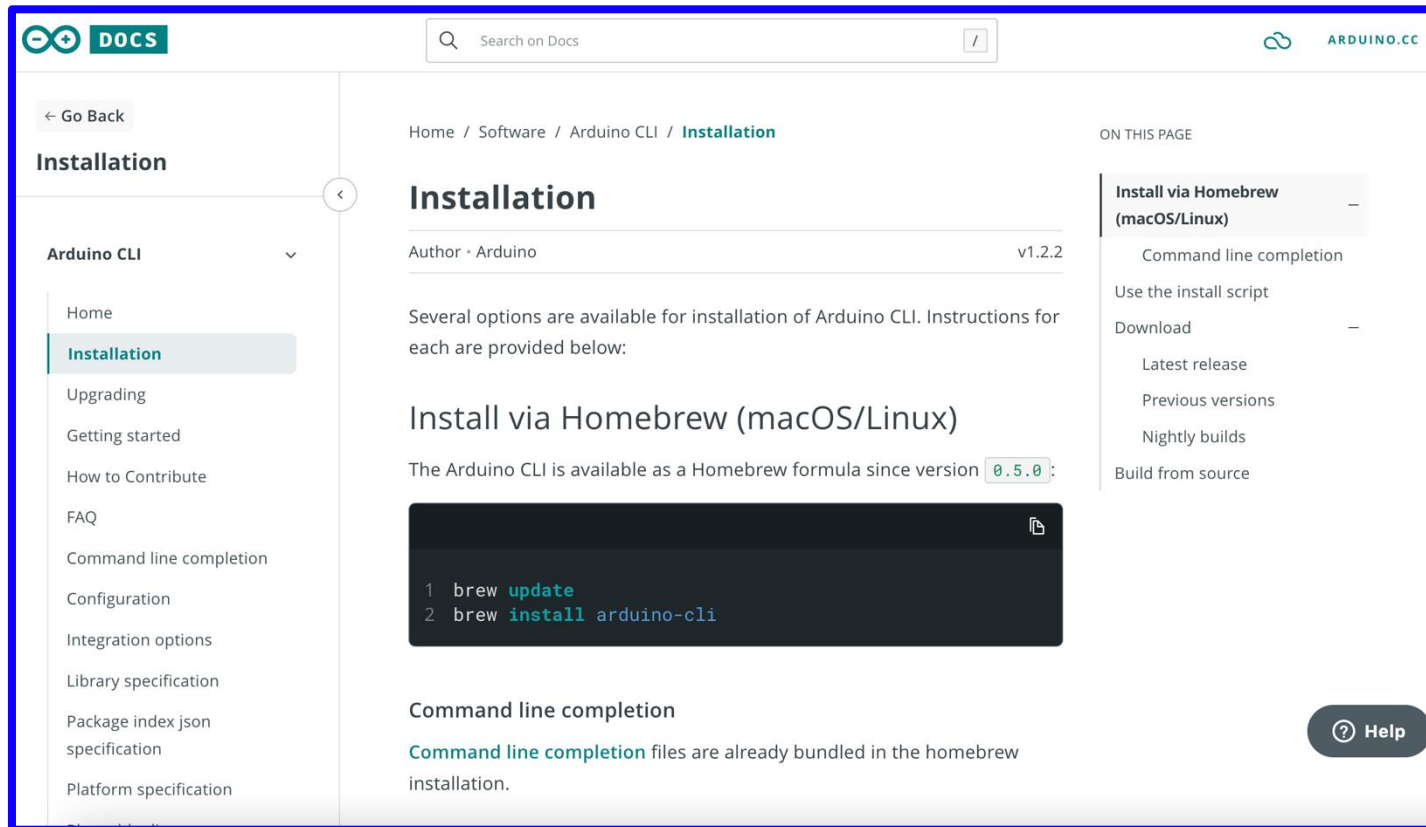
Try live inference preview on your device, from your terminal or command prompt run: `edge-impulse-run-impulse --debug`

Build



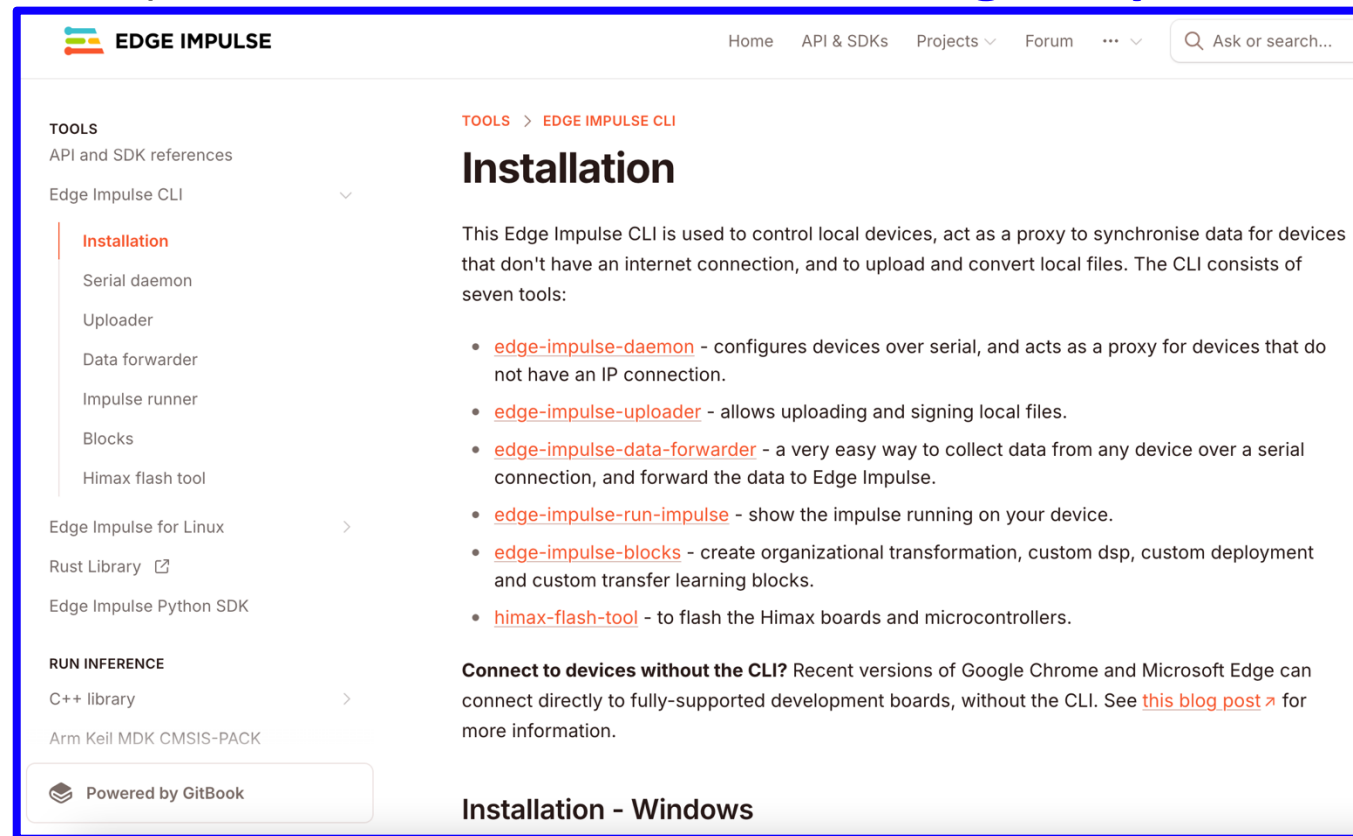
Lab 4 [Hardware] – Setting up Arduino CLI

- Go to <https://docs.arduino.cc/arduino-cli/installation/>
- **Carefully follow instructions** corresponding to your Laptop's/PC's operating system (Windows, MAC, Linux) for **Download and Install Arduino CLI**



Lab 4 [Hardware] – Setting up Edge Impulse CLI

- Go to <https://docs.edgeimpulse.com/docs/tools/edge-impulse-cli/cli-installation>
- **Carefully follow instructions** corresponding to your Laptop's/PC's **operating system** (Windows, MAC, Linux) for **Download and Install Edge Impulse CLI**



The screenshot shows the Edge Impulse CLI installation page. The left sidebar contains a navigation menu with sections: TOOLS (API and SDK references, Edge Impulse CLI, Installation, Serial daemon, Uploader, Data forwarder, Impulse runner, Blocks, Himax flash tool), Edge Impulse for Linux, Rust Library, Edge Impulse Python SDK, RUN INFERENCE (C++ library, Arm Keil MDK CMSIS-PACK), and a Powered by GitBook logo. The main content area is titled 'Installation' and describes the CLI's purpose. It lists seven tools: edge-impulse-daemon, edge-impulse-uploader, edge-impulse-data-forwarder, edge-impulse-run-impulse, edge-impulse-blocks, and himax-flash-tool. A section titled 'Connect to devices without the CLI?' provides information about connecting to development boards. The bottom of the page shows the 'Installation - Windows' section.

EDGE IMPULSE Home API & SDKs Projects Forum ... Ask or search...

TOOLS
API and SDK references
Edge Impulse CLI
Installation
Serial daemon
Uploader
Data forwarder
Impulse runner
Blocks
Himax flash tool

Edge Impulse for Linux
Rust Library
Edge Impulse Python SDK

RUN INFERENCE
C++ library
Arm Keil MDK CMSIS-PACK

Powered by GitBook

Installation

This Edge Impulse CLI is used to control local devices, act as a proxy to synchronise data for devices that don't have an internet connection, and to upload and convert local files. The CLI consists of seven tools:

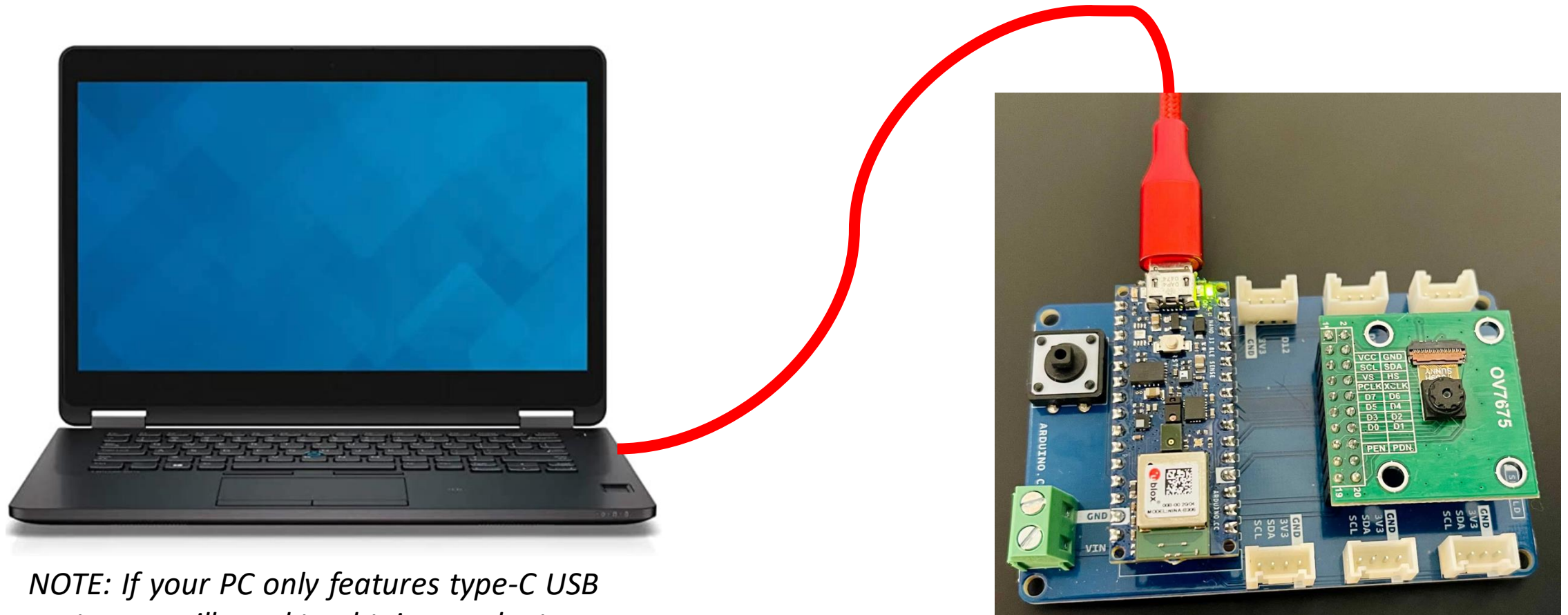
- **edge-impulse-daemon** - configures devices over serial, and acts as a proxy for devices that do not have an IP connection.
- **edge-impulse-uploader** - allows uploading and signing local files.
- **edge-impulse-data-forwarder** - a very easy way to collect data from any device over a serial connection, and forward the data to Edge Impulse.
- **edge-impulse-run-impulse** - show the impulse running on your device.
- **edge-impulse-blocks** - create organizational transformation, custom dsp, custom deployment and custom transfer learning blocks.
- **himax-flash-tool** - to flash the Himax boards and microcontrollers.

Connect to devices without the CLI? Recent versions of Google Chrome and Microsoft Edge can connect directly to fully-supported development boards, without the CLI. See [this blog post](#) for more information.

Installation - Windows

Lab 4 [Hardware] – Connect your TinyML kit

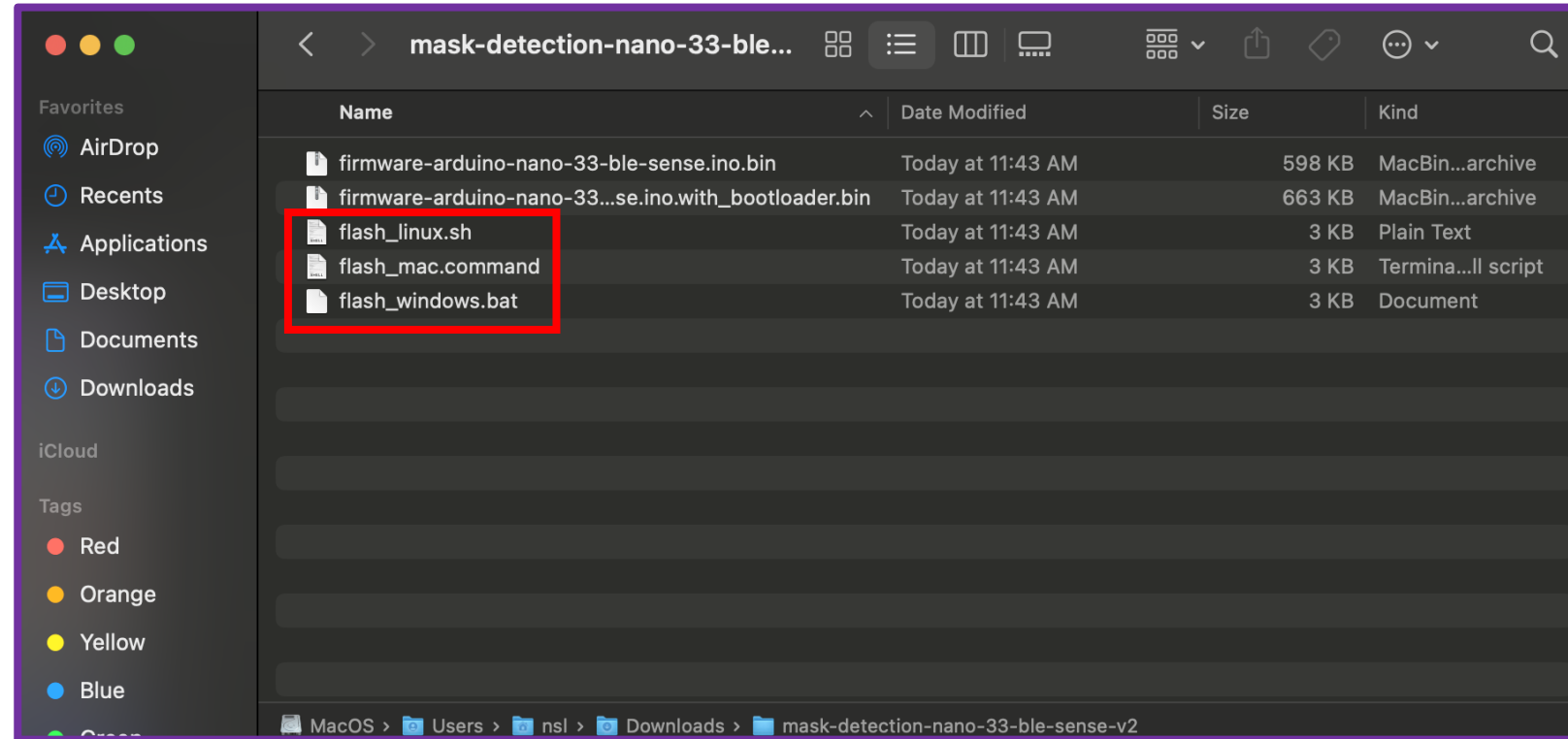
- Use the USB cable (type-A to microB) to connect the Nano 33 BLE Sense to your machine



NOTE: If your PC only features type-C USB ports, you will need to obtain an adaptor

Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- Flash the compressed model (**downloaded from Edge Impulse**) onto the board by **double clicking on the script for your operating system**



Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- **NOTE:** You may need to type the prompted command **“edge-impulse-run-impulse”** in a new command window

```
nsf — flash_mac.command — 80x24
Planes      : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 591376 bytes to flash (145 pages)
[=====] 100% (145/145 pages)
Done in 23.178 seconds
New upload port: /dev/cu.usbmodem143301 (serial)

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
logout

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

Lab 4 [Hardware] – Flash Your Model to TinyML Kit

- Follow the instructions prompted in command window to observe the results in the command line

```
ns1 — -bash — 80x24
Last login: Wed Jul  9 05:01:53 on ttvs000
ns1 — flash_mac.command — 80x24
Planes      : 1
Lock Regions : 0
Locked       : none
Security     : false
Erase flash

Done in 0.001 seconds
Write 591376 bytes to flash (145 pages)
[=====] 100% (145/145 pages)
Done in 23.178 seconds
New upload port: /dev/cu.usbmodem143301 (serial)

Flashed your Arduino Nano 33 BLE development board.
To set up your development with Edge Impulse, run 'edge-impulse-daemon'
To run your impulse on your development board, run 'edge-impulse-run-impulse'
logout

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

